Ernest Skrzypczyk

Master in Control Engineering, Robotics and Applied Informatics

6 Rue Mathurin Brissonneau, 220
44100 Nantes
France
✉ *ernest.skrzypczyk@gmail.com*
https://github.com/em-er-es/personal

Education

2015–2017 **Master in Control Engineering, Robotics and Applied Informatics – Advanced Robotics**, *Double Masters Degree*, *Erasmus Mundus*, European Masters in Advanced Robotics + (EMARO+), Università degli studi di Genoa / University of Genova, Italy – 1st year, École Centrale de Nantes / Centrale Nantes, France – 2nd year.

2007–2011 **Bachelor of Science - Electrical Engineer - Control Engineering and Robotics**, *Wrocław University of Technology*, Wybrzeże Wyspiańskiego 27, PL-50370 Wrocław, *very good / bardzo dobry*.
ISCED 5A, Polish grading system; See supplement to diploma, page 5

2004–2006 **Abitur**, *Berufsbildende Schulen Meppen - gew. und kaufm. Fachrichtungen - Fachrichtung Technik*, Nagelhof 83, D-49716 Meppen, *3,0 - good / gut.*
ISCED 3A, German grading system

Master thesis

Title *UGV and UAV collaboration within an autonomous infrastructure*

Supervisor Docent / Maître de conférences Olivier Kermorgant

Description Development of a collaborative, decentralized and heterogeneous SLAM based on previous work within ROS and V-REP framework at LS2N, formely known as IRCCYN.

Engineering project

Title *Metrological properties of a hybrid amplifier*

Supervisor PhD / dr inż. Daniel Dusza

Description Design, construction and measurements of a hybrid amplifier with a working frequency within the audible range and a microcontroller based control system.

Grade excellent / celujący

Prizes and Awards Stowarzyszenie Elektryków Polskich - Association of Polish Electrical Engineers
1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011

Experience

Vocational

11.2017 – present **Research engineer**, *IRT Jules Verne*, F-44340 Nantes/Bouguenais, France.
Vision and laser sensor data fusion, source code, mostly in C++, update, optimization and maintenance (CI), packaging, design and development of GUI for state representation as well as basic control of autonomous cobot unit *Asimov* working in workshop at Airbus.

01.02.2017 – 31.08.2017 **Master thesis work**, *École Centrale de Nantes, LS2N (formely known as IRCCYN) – ARMEN*, F-44300 Nantes, France.
Work on the master thesis as part of the ARMEN team.

2011 – 2015 **Self development**, *Improvement of skills in various areas, mostly hardware and GNU/Linux related*.

12.07.2010 – 20.08.2010 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein Berßen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems. Internship for studies at the Wrocław University of Technology.

14.07.2008 – 15.08.2008 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein Berßen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems.

2 weeks in May 2005 **Computer scientist**, *Ebert Mikrokomputer GmbH*, D-49716 Meppen, Germany.
Computer system installations mostly in the private sector. Customer care on site. Secondary school internship.

IRT Jules Verne

Sensor Pack **Objectives:**

- *Update, optimization and maintenance of developed source code*
  - *Target platform Ubuntu 16.06 / ROS Kane Kinetic*
  - *ROS best practices combined with in house conventions*
  - *Deployment of Ubuntu ROS packages*
- *Fusion of sensor data from several sources*
- *Integration of the sensor pack (SPACK)*
- *Design and implementation of GUI for state representation and simple control*
- *Continuous Integration (CI)*
- *Documentation of source code and ROS specific aspects*

Acquired experience:

- *Custom GIT workflow based on OneFlow*
- *Continuous Integration*
  - *Implementation of custom Continuous Integration on a local GitLab server*
  - *Implementation of Continuous Integration into the development process*
- *Dynamic Doxygen generated documentation*
- *Estimation of code performance in rudimentary benchmarks*

EMARO+

ECN **Projects and assignments:**
- *Homography based visual odometry*

- *Visual servoing and analysis using 2D, 3D and 2.5D/hybrid feature modelling*
- *Modelling and simulation of a 6 DOF articulated industrial robot in Delmia performing a spot welding task*
- *Optimal placement and kinematic design of a SCARA robot*
- *Kinematic design and analysis of a 3-RRR planer parallel robot*
- *Lateral and longitudinal control in a leader-follower and predefined path scenarios for a platoon of 5 vehicles*
- *Modelling, kinematic and dynamic analysis of a biglide mechanism, computed torque control (CTC) design and co-simulation in Adams and Matlab, crossing of type 2 singularity*
- *Modelling, kinematic and dynamic analysis of a 5-bar mechanism, CTC design and co-simulation in Adams and Matlab*
- *Modelling, analysis, experimentation and simulation of human motion using a motion capture system and pressure plate for different actor movements*
- *Dynamic modelling, passive walking, optimization and stability analysis of a compass robot*
- *Angular position and velocity estimation of a DC motor using Kalman filter*

UNIGE **Projects and assignments:**

- *Localization of a biped humanoid robot using EKF within ROS framework (**Github repository**)*
- *Study and implementation of developed SLAM algorithms for a biped humanoid robot within ROS framework (**Github repository**)*
- *Mobile robots control: point and posture tracking, Lyapunov control, design and simulation*
- *Mobile robots motion planning using bug, distance based and probabilistic roadmap planning algorithms*
- *Offline and online tuning of EKF in a real world case scenario using landmarks*
- *Technical report on Agriculture in controlled environments*
- *Technical paper review and presentation on FLEXA*
- *Delmia simulations using custom designed grippers and objects*
- *Design and evaluation of a user interface for a motion analysing application*
- *Basic image processing including edge and object detection, colour based segmentation, optical flow (Lucas-Kanade algorithm) and stereo vision*
- *POSIX compliant publisher / subscriber communication models using pipes and pthreads*
- *Realtime scheduling with RM, PS and EDF*

Languages

Polish	**Native**	
English	**B2+**	*Mark: very good / bardzo dobry at SJO WUT*
German	**C1**	*100% exam test score, 2nd place in contest at FLD WrUT*
Spanish	**A2**	*Mark: good, high school secondary language for 2 years*
Italian	**B1**	*Score 25/30, spent 1 year in Genova, Italy within EMARO+ programme*
French	**A1**	*Score 68/100, spent 1 year in Nantes, France within EMARO+ programme*

Computer skills

Operating systems	Extensive experience with Microsoft Windows, GNU/Linux, some experience with hardware specific solutions and FreeBSD	Distributions	Arch Linux (current main OS), OpenSUSE, Funtoo, Gentoo, Linux Mint, Ubuntu, Rasbian
Software experience	Various CAD/CAE, office and scientific applications and suites across most media fields including audio, video, graphics, text, design, simulation and engineering	Applications	LibreOffice, Microsoft Office, Altium Designer, Adams, Simatic Step 7, CX-Programmer, Delmia, Matlab, Spyder, Pythonxy, V-REP, Scilab, LabVIEW, Multisim, Proto, Photoshop, Gimp, Inkscape, Ableton Live, Audacity, Geany, TeXstudio, nano
Shells / Command line interfaces	Zsh (current main shell), Bash, PowerShell, Command Prompt	Software and community contributions	Proto, Arch Linux Wiki articles (ALSA - Basic Syntax), Arch Linux Forums and packages maintenance, wine, q4wine

Repositories

GitHub	http://github.com/em-er-es	Scripts	http://github.com/em-er-es/scripts
Robotics	http://github.com/em-er-es/rollo http://github.com/em-er-es/rollo-slam	EMARO+	http://github.com/em-er-es/reports

Programming languages

General languages

- *Assembler on various hardware*
- *BASIC on Commodore 64 and various BASIC derivatives on robots*
- *C/C++ in courses, assignments, projects and contributions*
- *Python*
- *LUA*

Frameworks

- *ROS - Robot Operating System*
- *V-REP*

PLC languages

- *Function block diagram*
- *Ladder logic*

Scripting

- *Bash / Zsh*
- *Matlab / Scilab / Pythonxy / Spyder*
- *Command Prompt / Batch*
- *Autohotkey*

Hardware

Electrical	Electrical motors, industry grade electrical motors, [car] audio equipment
Electronical	Embedded systems (8051, AVR ATtiny & ATmega)
Robotics	Biped humanoid robot, industrial articulated robot
Skills	Soldering THD and SMD, basic material processing

Certificates, Licences

Electrical qualification	**Operation position** in the area of: **operation**, **maintenance**, **repairs** and **assembly**	*SEP - AoPEE, Wrocław* *Date of exam: 19.05.2010* *Valid for 5 years*
Driving licence	**Class A, B**	*Driving licence acquired in Germany in 2005*

Organisations, associations and memberships

ECN	Student representative for EMARO+ ECN	
SNS STRIMER	Academic student association Wrocław University of Technology	*Took part in several projects, educational trip and made a publication for the SSC 2009*

Additional courses

	Nantes
2017	Autonomous vehicles
2017	Humanoid and walking robots
	Warsaw
2015	ABC of enterneurship
2015	Self presentation and interview skills

References

Please contact me for specific information

Publications

Rzepecki Bartłomiej, Skrzypczyk Ernest, "*Plasma and its applications*", 7th Students' Science Conference, Wrocław, Poland, 2009

Prizes and Awards

Prize	1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011
Prize	Second place in German language tournament at FLD WrUT
Commendation	For best paper presentation at SSC 2009

Aknowledgment Third place for best group work at SSC 2009

Scholarships

Several scholarships for achievements in studies at the WrUT Faculty of EE

Interests

Arts
- *Graphics, photography, image rendering and manipulation*
- *Music, audio creation and manipulation*
- *Paper model building*

Philosophy
- *New socio-economic structures*
- *Consequences of implementation of new technologies*

Technology
- *Self-sustainable autonomous vehicles, facilities and infrastructure*
- *Robot interaction*
- *Artificial intelligence*
- *FOSS, especially GNU/Linux*
- *Open Hardware, Open Technology, Open Economy*

Additional information

Mobility Full mobility within 157 countries, especially across Europe

Self-development Main focus was to shift working environment to and towards FOSS, contribute and exchange with the FOSS community.